DiffuGC: Diffusion Model Can Help Discover Granger Causality from Interventional Time Series

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2025 IEEE International Conference on Data Mining (ICDM) | 979-8-3315-9599-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDM65498.2025.00056

***Abstract*—Discovering Granger causality from time series data is fundamental to understanding dynamic systems, yet most existing methods struggle with unknown intervention targets or causal structures in real-world scenarios. In this paper, we propose DiffuGC, a novel diffusion-based framework that unifies observational and interventional causal discovery through a generative denoising process. By introducing diffusive interventions, which apply progressive interventions without any prior knowledge, DiffuGC amplifies causal signals while preserving structural information. Furthermore, we introduce a denoising NoiFormer with adaptive attention to both short- and long-term causal dependencies, which disentangles trend and seasonal components to enable accurate reconstruction of causal structures from interventional data. To the best of our knowledge, we are the first to integrate diffusion models with interventional Granger causal discovery. Extensive experiments on synthetic, quasi-real, and real-world benchmarks demonstrate that DiffuGC consistently outperforms state-of-the-art baselines in both observational and interventional data. Moreover, we introduce an intriguing notion, Causality Acceleration, characterized by the early emergence of informative causal patterns within the diffusion path, which may open up promising directions for future research on efficient and adaptive causal discovery.**

***Index Terms*—Granger causal discovery, Diffusion models, Observational and interventional time series, Diffusive intervention**

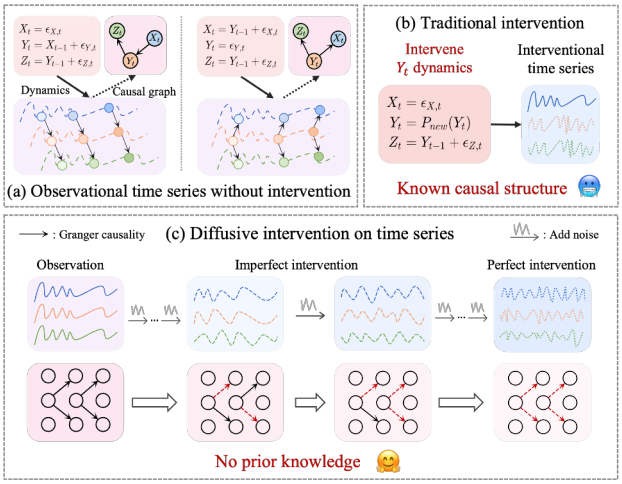


Fig. 1: Integrating diffusion models into causal discovery addresses key challenges: (a) observational data cannot uniquely identify causal structures; (b) traditional interventions require known targets or causal graphs; (c) diffusion enables progressive, prior-free intervention, facilitating causal inference.

I. Introduction

Causal interpretation of the observed time-series data can help answer fundamental causal questions and advance scientific discoveries in various disciplines such as medical and financial fields [1], [2]. Researchers in the past decades have been dedicated to discovering causal graphs from observed time-series and made large progress, for example, Granger causality [3], transfer entropy (TE) [5], constraint-based methods (e.g., PCMCI [6]), noise-based methods (e.g., VarLiNGAM [4]). Unfortunately, conditional TE and constraint-based algorithms struggle to scale due to their iterative procedures and reliance on density estimation, highlighting the need for more efficient techniques in discovering causal relationships. Recently, deep learning methods have shown strong potential in Granger causal discovery [2], [7], [8], demonstrating effectiveness in modeling pairwise interactions in domains like gene regulatory networks [2], [38].

Despite recent progress, most existing methods infer causal graphs from purely observational time series. However, learning causal structures solely from observational data remains fundamentally limited. Under the faithfulness assumption, the true causal structure is only identifiable up to a Markov Equivalence Class (MEC) [12], where multiple distinct graphs yield identical observational distributions [10], [11]. As illustrated in Fig. 1, even structurally different causal graphs can produce indistinguishable time series when noise terms ϵ_t are independent and Gaussian. Encouragingly, this identifiability issue can be alleviated by incorporating interventional data, which introduces distribution shifts across different conditions. Such shifts present both challenges and opportunities for causal discovery: under interventions, the causal structure becomes identifiable within a stricter Interventional MEC (I-MEC) [13], [14]. With sufficiently diverse interventions, even full recovery of the true causal graph becomes theoretically possible [15].

While some prior studies have explored causal discovery from interventional time series, they often rely on strong assumptions—such as known intervention targets [13], [18] or

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access to ground-truth causal structures to guide the intervention process [17]. However, such assumptions rarely hold in real-world settings. Interventions are often imperfect and their targets unknown, as in cases involving environmental shifts or system malfunctions [17]. Furthermore, real-world time series frequently exhibit heterogeneity: observational segments, perfect interventions, and imperfect interventions may coexist within the same dataset. This diversity substantially increases the complexity of causal discovery under interventional settings. Consequently, successfully deciphering causal structures within time series that incorporate both observational and interventional data, especially when interventions are unknown or imperfect, continues to be a crucial and predominantly unresolved challenge.

To address these challenges, we propose DiffuGC, a novel framework for Granger causal discovery that integrates forward diffusive intervene process with reverse causal discovery mechanism. To the best of our knowledge, DiffuGC is the first to incorporate Denoising Diffusion Probabilistic Models (DDPMs) [16] into the context of interventional Granger causality. By progressively injecting noise in the forward diffusion process, DiffuGC constructs a natural continuum from observational time series to imperfect and ultimately perfect interventions—without requiring any prior knowledge of intervention targets. To reconstruct causal structures from such heterogeneous data, we introduce a denoising NoiFormer with adaptive attention to both short- and long-term causal dependencies, which disentangles trend and seasonal components to enable accurate reconstruction of causal structures from interventional data. Throughout the full diffusion trajectory, DiffuGC learns to reconstruct clean causal patterns from time series subjected to varying degrees of intervention, ranging from none to complete. This enables DiffuGC to handle real-world scenarios involving complex, unknown, and imperfect interventions in a unified and fully data-driven manner. Code of DiffuGC is avaliable and anonymous at https://anonymous.4open.science/r/DiffusionGC. In summary, the main contributions of this paper are as follows:

- We introduce DiffuGC, a novel framework that combines forward diffusive intervention with reverse causal discovery. DiffuGC is the first to incorporate diffusion models into interventional Granger causal discovery. By leveraging DDPMs, it generates intervention-aware time series without requiring any prior knowledge, even in complex scenarios involving both observational and interventional data, and enables step-wise learning of causal features throughout the diffusion trajectory.
- We develop NoiFormer, a Transformer-based denoiser that adaptively models both short- and long-term causal dependencies. It disentangles trend and seasonal components from noisy sequences, enabling causal graph reconstruction under varying degrees of intervention.
- Extensive experiments on synthetic, quasi-real, and real-world datasets show that DiffuGC consistently outperforms state-of-the-art baselines under both observational and interventional conditions.
- We introduce the notion of Causality Acceleration, suggesting new directions for efficient and adaptive causal discovery.

II. Related Work

A. Causal Discovery from Intervention Data

In parametric analyses, datasets from various distributions, or domains, are considered interventional data. MC/IB [19] explored causal graph learning in linear systems using observational data across domains with varying causal coefficients. DCDI [13] proposed a differentiable causal graph learning method for static data using score functions to manage perfect, imperfect, and unknown interventions, identifying the I-MEC. Recent advances integrate large language models with score-based temporal causal discovery [20] to create intervention-aware scoring via semantic pattern recognition. Many intervention-based causal discovery methods depend on scores, limiting scalability and expressiveness. However, Granger causality, compatible with neural networks, promises scalable and dynamic causal inference.

B. Neural Granger Causal Discovery

Granger causality [3] identifies causality in time-series: if past data of variable A improves predictions of variable B, then A Granger causes B. Deep learning integration with Granger causality is promising for enhancing causal modeling. NGC [2] introduced a sparse-input MLP and LSTM for nonlinear Granger causality. eSRUS [9] proposed a recurrent unit with groupwise regularized inputs. CR-VAE [8] incorporates Granger causality using a causal variational autoencoder. CUTS [7] created a method for a causal adjacency matrix in high-dimensional data with sparse regularization. These methods infer Granger causal graphs but don't fully leverage interventional data or solve identifiability issues. Integrating Granger causality with interventional analysis is mostly limited to IGC [17], which relies on impractical assumptions and prior causal knowledge, highlighting gaps in identifying Granger causality in observational data with unknown interventions and hidden mechanisms.

C. Denoising Diffusion Probabilistic Models

Denoising diffusion probabilistic models (DDPMs) are innovative generative frameworks [16]. These models gradually convert noise to structured data through a denoising process guided by learned reverse transitions. Diffusion-based methods excel in time series forecasting, capturing uncertainty and complex temporal dependencies [21], [22]. Recent research explores integrating DDPMs with causal reasoning by creating causal variants [24], [25]. However, these studies mainly simulate rather than discover causal mechanisms. Research on using diffusion models for causal discovery is limited [26], [27], and interventional Granger causal discovery with DDPMs remains unexplored.

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III. Preliminaries

A. Problem Formualtion

Granger Causality Discovering: Consider a complex dynamical system modeled by a multivariate time series $\mathbf{X} = \{X_1, \ldots, X_d\} \in \mathbb{R}^{D \times T}$ with D variables. The system's dynamics are recorded over T observation points, $t \in (1, \ldots, T)$. The causal relationships between variables are described by this structural model:

$$X_{i,t+1} = g_i(X_{1,<t}, ..., X_{D,<t}) + \epsilon_{l,n} \tag{1}$$

where $X_{j,<t} = (X_{j,1}, \ldots, X_{j,t-1})$ denotes the past of time series i and $g_i(\cdot)$ is a function mapping the past of all the D time series to series i.

Time series j is Granger non-causal for time series i if for all $X_{1,<t}, ..., X_{D,<t}$ and all $X'_{j,<t} \neq X_{j,<t}$:

$$\begin{aligned} &g_i(X_{1,<t}, \ldots, X_{j,<t}, \ldots, X_{D,<t}) \\ =&g_i(X_{1,<t}, \ldots, X'_{j,<t}, \ldots, X_{D,<t}) \end{aligned} \tag{2}$$

which implying $g_i(\cdot)$ does not depend on $X_{j,<t}$. Often the Equation 2 is reflected on the parameters of neural networks.
Causal Graph Construction: Let $G = (V, E)$ be the Granger causal graph, with V as nodes and E as edges. Set V includes D dependent time series $(X_1, \ldots, X_d)$. An edge from node X_i to X_j exists if: 1) $i \neq j$ values of X_i give unique, significant predictive information about X_j; and 2) for $i = j$, X_i self-causes X_i.

B. Intervention

An intervention on a variable involves replacing its conditional density with a new one. We also define the **interventional targets**, a set of variables subjected to simultaneous intervention, and the **interventional family** $\mathcal{I} := (I_1, \ldots, I_{\mathcal{O}})$, where $\mathcal{O}$ counts the interventions. The observation context, with no intervention, is denoted by $I_1 := \emptyset$. The likelihood for the kth intervention is expressed as:

$$p^{(o)}(\mathbf{X}) := \prod_{j \notin I_o} p_j^{(1)}(X_j|\text{PA}(X_j)) \prod_{j \in I_o} p_j^{(o)}(X_j|\text{PA}(X_j)) \tag{3}$$

Interventions are of two types: 1) imperfect (or soft) interventions [23] and 2) perfect (or hard) interventions [13], which disconnect a node from its parent nodes $j \in I_o$ $p_j^{(o)}(X_j|\text{PA}(X_j)) = p_j^{(o)}(X_j)$. Typically, intervention methods assume known targets, limiting Granger causal discovery when targets are unknown or cannot be predetermined.

IV. DiffuGC

In this section, we introduce DiffuGC mainly consists of DDPMs, in which Granger causal discovery is embedded, as shown in Fig.2.

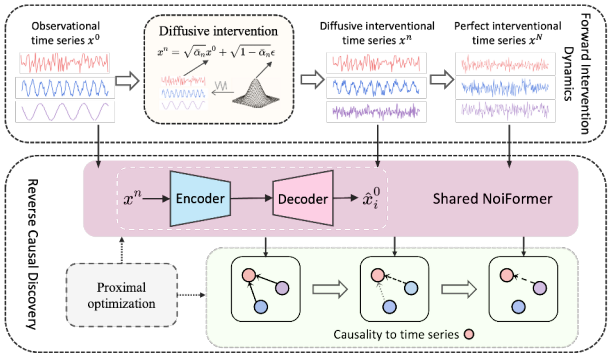


Fig. 2: Illustration of the proposed DiffuGC architecture.

A. Forward Intervention Dynamics: Diffusive Interventions on Time Series

Within the framework of the diffusion model, the forward intervention process q is a Markov process in which Gaussian noise is gradually added to the time series at each step until the original data $x^0 \sim q(x^0)$ transforms into complete Gaussian noise x^N, following an increasing variance schedule $\beta_1, \ldots, \beta_N$ with $\beta_n \in (0, 1)$ a variance at diffusion step n. The approximate posterior $q(x^{<N}|x^0)$ is then computed as follows:

$$q(x^{<N}|x^0) = \prod_{n=1}^{N} q(x^n|x^{n-1}) \tag{4}$$

$$q(x^n|x^{n-1}) = \mathcal{N}(x^n; \sqrt{1-\beta_n}x^{n-1}, \beta_n\mathbf{I}) \tag{5}$$

where N denotes the total number of steps in the forward process. The forward process variance β^n in (2) can be implemented with the reparameterization trick:

$$x^n = \sqrt{1-\beta_n}x^{n-1} + \sqrt{\beta_n}\epsilon, \tag{6}$$

where $\epsilon \sim \mathcal{N}(0, I)$.

A notable property of the forward process is that using notation $\alpha_n = 1 - \beta_n$ and $\sqrt{\bar{\alpha}_n} = \prod_{s=1}^{n} \alpha_s$, we can sample x^t at any arbitrary time step t in a closed form:

$$q(x^n|x^0) = \mathcal{N}(x^n; \sqrt{\bar{\alpha}_n}x^0, (1-\bar{\alpha}_n)\mathbf{I}) \tag{7}$$

$$x^n = \sqrt{\bar{\alpha}_n}x^0 + \sqrt{1-\bar{\alpha}_n}\epsilon \tag{8}$$

Intermediate noised data points x^n represent partially intervened samples, which transform into fully intervened time series after the final step N, x^n. This transition helps the model capture various intervention intensities, enriching causal discovery and boosting its ability to identify intervention-aware causal relationships.

B. Reverse Causal Discovery: Denoising on Heterogeneous Time Series

The DDPMs' reverse process is a Markov process that restores data by incrementally removing noise from corrupted

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time series $x^0 x^N$. A neural network learns to denoise samples through this reverse transition θ:

$$p_\theta(x^{n-1}|x^n) = \mathcal{N}(x^{n-1}; \mu_\theta(x^n, n), \tilde{\beta}_n \mathbf{I}) \tag{9}$$

where $\mu_\theta(x^n, n)$ is the mean of reverse distribution obtained from neural network θ, $\tilde{\beta}_n = \frac{1-\bar{\alpha}_{n-1}}{1-\bar{\alpha}_n}\beta_n$.

Inspired by [21], we directly predict the original time series x^0 from each intermediate state x^n in the forward diffusion process, enabling the extraction of Granger causal relationships from interventional data. Specifically, for each time series i, we construct a dedicated fitting function f_{θ_i} to model its causal dependencies under the dynamics of diffusion:

$$\hat{x}_i^0 = f_{\theta_i}(x^n, n) \tag{10}$$

As the forward interventional process unfolds, the function f_{θ_i} progressively learns from each intermediate diffused state to approximate the underlying dynamics of time series variable i. Building upon Eq. 10, Granger causality is ultimately determined based on the following criterion for all $x'^n_j \neq x^n_j$:

$$f_{\theta_i}((x^n_1, \ldots, x^n_j, \ldots, x^n_d), n) = f_{\theta_i}((x^n_1, \ldots, x'^n_j, \ldots, x^n_d), n) \tag{11}$$

C. Shared Denoing NoiFormer

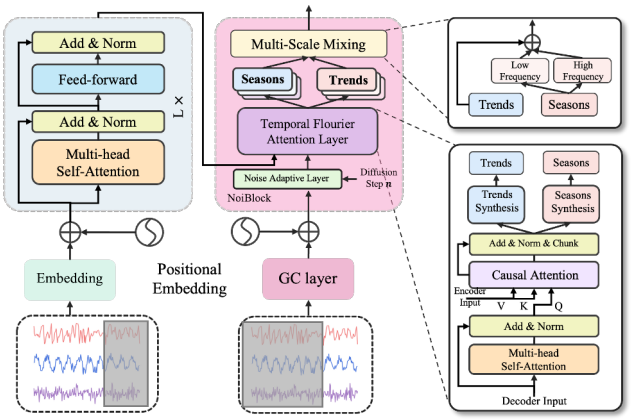


Fig. 3: The network architecture of NoiFormer.

We introduce NoiFormer, a Transformer-based encoder-decoder model, to extract dynamics from noise-corrupted sequences (Fig. 3). NoiFormer decomposes inputs into trends, seasonality, and error to better model noisy temporal patterns. Trends depict smooth changes, seasonality captures periodic patterns, and errors are the residuals. For Granger causality, the time series passes through a GC layer for embedding, where the weight matrix W^0 indicates the causal impact of D on target i. NoiFormer's attention layers then model features and extract causal information.

$$\mathbf{h}_1^t = W^0 x^{n,(t_s,\ldots,t_e)} + b^0 + PE_{pos} \tag{12}$$

$$w^{l,n}_{(tr)}, w^{l,n}_{(seas)} = \text{TFA}(\mathbf{h}_1^t, x^{n,(t_e,\ldots,t-1)}) \tag{13}$$

where $x_i^{n,t}$ is the value of the interventional time series i at time t and diffusion step n. $(t_s, \ldots, t-1)$ is the prediction look-back window, with its earlier segment $(t_s, \ldots, t_e)$ as the encoder input and the latter segment $(t_e, \ldots, t-1)$ as the decoder input. TFA (Temporal Fourier Attention layer) integrates encoder and decoder outputs using self and cross-attention to model trends, seasonality, and residuals in time series. $w^{l,n}_{(\cdot)}, i \in (1, \cdots, L)$ indicate the relevant decoder block index at diffusion step n.

Trend Synthesis. The trend component describes the smooth underlying mean of the data, which aims to model slow-varying behavior. To produce reasonable trend components, we use the polynomial regressor to model the trend V_{tr}^n as follows:

$$V_{tr}^n = \sum_{i=1}^{D} (\boldsymbol{C} \cdot \text{Linear}(w_{tr}^{l,n}) + \mathcal{X}_{tr}^{l,n}), \quad \boldsymbol{C} = [1, c, \ldots, c^s] \tag{14}$$

where $\mathcal{X}_{tr}^{l,n}$ is the mean value of the output of the ith decoder block, and '$\cdot$' denotes tensor multiplication. Here slow-varying poly space C is the matrix of powers of vector $c = [0, 1, 2, \cdots, \tau-2, \tau-1]T/\tau$, and s is a small degree to model low frequency behavior.

Seasonality & Error Synthesis. We aim to extract non-trend components from the model input, including seasonal and error elements, by representing seasonality with Fourier synthetic layers and bases. We also observe that at larger diffusion steps, the intervened time series tends to preserve long-term trends and variations, which correspond to low-frequency components in the Fourier domain, reflecting long-term causal effects. In contrast, at smaller diffusion steps, the sequences contain less noise and retain more short-term fluctuations, associated with high-frequency components and short-term causal effects. Motivated by this insight, we propose assigning noise-aware frequency weights to high- and low-frequency channels post-Fourier transform, enabling the model to adaptively focus on the most informative frequency components under different noise levels.

$$A_{l,n}^{(k)} = \left|\mathcal{F}(w_{seas}^{l,n})_k\right| \tag{15}$$

$$\Phi_{l,n}^{(k)} = \phi\left(\mathcal{F}(w_{seas}^{l,n})_k\right) \tag{16}$$

$$\kappa_{l,n}{}^{(1)}, \cdots, \kappa_{l,n}^{(K)} = \underset{k\in\{1,\cdots,\lfloor\tau/2\rfloor+1\}}{\arg\,\text{TopK}} \{A_{l,n}^{(k)}\} \tag{17}$$

$$\kappa_{l,n}^{LF(1)}, \cdots, \kappa_{l,n}^{LF(K_{LF})} = \underset{k\in\kappa_{l,n}{}^{(1)},\cdots,\kappa_{l,n}^{(K)}}{\arg}(\Phi_{l,n}^{(k)} < \max(\Phi_{l,n})/2) \tag{18}$$

$$\kappa_{l,n}^{HF(1)}, \cdots, \kappa_{l,n}^{HF(K_{HF})} = \underset{k\in\kappa_{l,n}{}^{(1)},\cdots,\kappa_{l,n}^{(K)}}{\arg}(\Phi_{l,n}^{(k)} > \max(\Phi_{l,n})/2) \tag{19}$$

$$S_{l,n}^{LF} = \sum_{k=1}^{K_{LF}} A_{l,n}^{\kappa_{l,n,LF}^{(k)}} [\cos(2\pi f_{\kappa_{l,n,LF}^{(k)}} \tau c + \Phi_{l,n}^{\kappa_{l,n,LF}^{(k)}}) + \cos(2\pi \bar{f}_{\kappa_{l,n,LF}^{(k)}} \tau c + \bar{\Phi}_{l,n}^{\kappa_{l,n,LF}^{(k)}})] \tag{20}$$

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$$S_{l,n}^{HF} = \sum_{k=1}^{K_{HF}} A_{l,n}^{\kappa_{l,n,HF}^{(k)}} [\cos(2\pi f_{\kappa_{l,n,HF}^{(k)}} \tau c + \Phi_{l,n}^{\kappa_{l,n,HF}^{(k)}}) + \cos(2\pi \bar{f}_{\kappa_{l,n,HF}^{(k)}} \tau c + \bar{\Phi}_{l,n}^{\kappa_{l,n,HF}^{(k)}})] \tag{21}$$

where the function TopK is utilized to extract the top K amplitudes, where K is considered a hyperparameter. The terms $A_{l,n}^{(k)}$ and $\Phi_{l,n}^{(k)}$ denote the amplitude and phase of the k-th frequency component after applying the discrete Fourier transform, $\mathcal{F}$. The symbol f_k signifies the Fourier frequency associated with index k, and $\bar{(\cdot)}$ indicates the conjugate of $(\cdot)$.

Avoiding the introduction of additional hyperparameters, we use $\sqrt{\bar{\alpha}_n}$ as the adaptive noise-aware weighting factor, enabling the model to dynamically adjust its sensitivity to the noise level at each diffusion step. Ultimately, the original signal can be reconstructed using the following equations:

$$\hat{x}^0(x^n, n, \theta) = V_{tr}^n + (1-\sqrt{\bar{\alpha}_n}) \sum_{l=1}^{L} S_{l,n}^{LF} + (\sqrt{\bar{\alpha}_n}) \sum_{l=1}^{L} S_{l,n}^{HF} + R \tag{22}$$

where R represents the output from the final decoder block and can be considered as the combined result of residual periodicity along with other noise.

D. Granger Casual Disvocery

While modeling target variable i, NoiFormer learns from each diffusion step, and the GC layer updates causal weights from time series D to variable i. We apply a regularization term on W^0 to the training loss to promote sparsity in the causal matrix W^0, improving interpretability.

$$\mathcal{L} = \sum_{t=2}^{T} \underbrace{(\hat{x}_i^{0,t} - f_{\theta_i}(x^{n,(t_e,\cdots,t-1)}, n)^2)}_{\mathcal{L}_{pred}} + \underbrace{\lambda \sum_{j=1}^{D} \| W_{:j}^0 \|_2}_{\mathcal{L}_{penalty}} \tag{23}$$

1) Optimizing the Penalized Objective: We use proximal gradient descent [33] to optimize the nonconvex objectives of Eq. 23. This approach is crucial in inducing zeros in input matrix columns, key for interpreting Granger non-causality. A line search might be added to the algorithm to ensure local minimum convergence [34]. The algorithm updates the network weights $\boldsymbol{\theta}_i$ iteratively starting with $W^{(0)}$ by

$$W^0(i+1) = \text{prox}_\gamma \left(W^0(i) - \gamma \nabla \mathcal{L}_{pred}(W^0(i)) \right) \tag{24}$$

where prox_γ is the proximal operator with step size γ; $\mathcal{L}_{pred}$ is the convex part of $\mathcal{L}$.

The weights' group lasso penalty proximal step involves a group soft-thresholding operation [33]:

$$\text{prox}_{\gamma\lambda}(W_{:j}^0) = \text{soft}(W_{:j}^0, \gamma\lambda) = \left(1 - \frac{\lambda\gamma}{||W_{:j}^0||_2}\right)_+ W_{:j}^0 \tag{25}$$

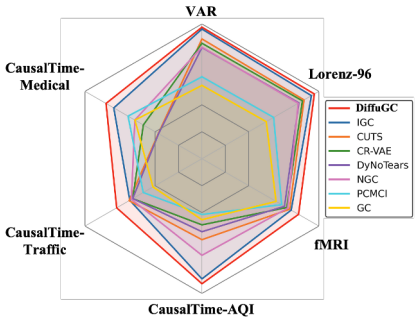


Fig. 4: Summary of DiffuGC's performance across six benchmark datasets

where $(x)_+ = \max(0, x)$. Training uses two optimization methods: proximal gradient on input layer weights $W^{(0)}$, and SGD on other parameters.

Algorithm 1 Proximal gradient descent with line search algorithm.

Require: $\lambda > 0$
  $m = 0$, initialize $\Theta, W^0(0)$
  while not converged **do**
    compute $\nabla \mathcal{L}_{pred}$ by BPTT and pdate Θ except W^0 using SGD.
    $i = i + 1$
    determine γ by line search.
    for $j = 1$ to m **do**
      $W_{:j}^0(i+1) = \text{soft}\left(W_{:j}^0(i) - \gamma \nabla_{W_{:j}^0} \mathcal{L}_{pred}\left(\mathbf{W}^0(i)\right), \gamma\lambda\right)$
    end for
  end while
  return $(\mathbf{W}^0(i))$

2) Model Complexity: DiffuGC's complexity arises from d-fold forward intervention and backward causal discovery. Forward diffusion complexity is $\mathcal{O}(Nd)$ over N steps. In the backward phase, a NoiFormer module with L layers processes a time series of length T and hidden dimension H, with complexity $\mathcal{O}(LT^2H)$. All d instances execute independently in parallel, achieving a parallel-time complexity of $\mathcal{O}(Nd + LT^2H)$, enabling scalability for large causal graphs.

V. Experiments

We evaluate DiffuGC's performance using two methods: (1) using only observational time series data, and (2) combining observational and interventional time series data. Figure 4 summarizes its performance on six benchmark datasets. We also apply DiffuGC to a real-world causal discovery task, aligning with established medical research findings and showcasing its practical effectiveness.

A. Datasets

We tested our approach on 2 synthetic (VAR, Lorenz-96) and 4 public quasi-real datasets (fMRI, Air Quality Index,

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Traffic, Medical). The synthetic datasets are observational, while the quasi-real datasets are interventional due to modifications embedding known causal structures.

- VAR (linear): As a linear dataset, VAR datasets are simulated following:

$$x^t = \sum_{\tau=1}^{\tau_{max}} \mathbf{A}_\tau x^{t-\tau} + e_t, \tag{26}$$

  where the matrix $\mathbf{A}_\tau$ is the sparse autoregressive coefficients for time lag τ. Time-series i Granger cause time-series j if $\exists \tau \in \{1, ..., \tau_{max}\}, a_{\tau,ij} > 0$.
- Lorenz-96 (non-linear): As a nonlinear model to simulate climate dynamics, the p-dimensional Lorenz-96 [30] model is defined as:

$$\frac{dx_i^n}{dt} = (\mathbf{x}^{i+1} - \mathbf{x}^{i-2})\mathbf{x}^{i-1} - \mathbf{x}^i + F \tag{27}$$

  where $x_{-1} = x_{p-1}$,$x^0 = x_p$ and $x_{p+1} = x_1$. F is the forcing constant which is set to be 10.
- fMRI: fMRI [29] serves as a benchmark for causal discovery, comprising realistic simulations of blood-oxygen-level-dependent (BOLD) time series generated via the dynamic causal modelling framework for functional magnetic resonance imaging (fMRI) forward model.
- Air Quality Index, Traffic and Medical: The CausalTime [28] pipeline, serving as a novel benchmark dataset, produces time series that closely mimic real-world data, incorporating ground truth causal graphs for quantitative performance assessment. The CausalTime benchmark encompasses three domains of quasi-real time series: Air Quality Index, Traffic, and Medical.

B. Baselines and Metrics

1) Baselines: **GC** [3] (Vector AutoRegressive) is a linear model for Granger causality tests. **PCMCI** [6] uses conditional independence tests with optimized conditioning sets for inferring causal structure. **NGC** [2] implements a component-wise LSTM with sparse input weights to infer non-linear Granger causality. **DyNoTears** [31] is a score-based method with continuous optimization for learning causal structure. **CR-VAE** [8] are specialized RNNs for identifying the structure of non-linear Granger causal networks. **CUTS** [7] is a neural Granger causal discovery algorithm for imputed and high-dimensional data. **IGC** [17] combines intervention modeling and Granger causality to robustly infer multivariate TS causal structures, accommodating distribution shifts or interventions.

2) Evaluation Metrics: We use four standard evaluation metrics: **AUROC** Area Under the Receiver Operating Characteristic curve shows the area under the true positive rate vs. false positive rate curve across thresholds; **AUPRC** Area Under the Precision-Recall Curve analyzes precision vs. recall at various thresholds; **F1 Score** measures the harmonic mean of precision and recall; **SHD** Structural Hamming Distance counts incorrectly predicted edge states.

C. Implementation Details

We used the official implementations of baseline methods with parameter adjustments for our setup. Hyperparameter search was conducted over batch size $b \in 16, 32, 64, 128$, learning rate $lr \in (5 \times 10^{-4}, 1 \times 10^{-4})$, weight coefficient $\lambda \in (1 \times 10^{-1}, 1 \times 10^{-2})$, and proximal step size $\gamma \in 1, 5, 10, 20$. Diffusion models were tested with a noise level of $N = 1000$.

D. Experiment Results

In this section, we evaluate the effectiveness of DiffuGC in discovering Granger causalit. We highlight the best and the second best in bold and with underlining, respectively.

VAR. We simulated $D = 10, 20, 40$ time series over $T \in (500, 1000)$ observations with a maximum time lag of $\tau \in 5, 20$. Table I shows DiffuGC's performance on VAR datasets across different dimensionalities, lengths, and lags. DiffuGC consistently achieves the highest AUROC, AUPRC, and F1 scores, with the lowest SHD. Even in challenging conditions (D=30, Lag=20), DiffuGC outperforms GC and TCDF, which degrade significantly. Compared to neural methods like NGC, CR-VAE, and CUTS, DiffuGC shows stronger generalization and better causal structure recovery. It also demonstrates more robustness than IGC, with less variability and consistently lower SHD, confirming its reliability in identifying causal relations in complex scenarios.

Lorenz-96. For this dataset, we simulated $T \in (500, 1000)$ observations with $D \in (20, 40)$, $F \in 10, 20$. F is a forcing constant, and higher F values increase chaotic behavior and nonlinearity. Table II shows DiffuGC's performance on the Lorenz-96 system across different dimensions and chaos levels. DiffuGC consistently outperforms existing methods. It excels in both low-dimensional, mildly nonlinear and high-dimensional, highly chaotic scenarios, showing superior accuracy and stability in structural recovery with low SHD values. Unlike others, DiffuGC maintains robust performance under strong chaos, demonstrating strong adaptability and generalization in complex nonlinear systems.

fMRI. The simulated fMRI BOLD signals, generated using DCM with a nonlinear balloon model, include time series for various brain ROIs across 28 sub-datasets, each with 50 unique-subject features. This study evaluates all subjects and simulations, unlike previous limited studies. Table III shows that DiffuGC excelled in 22 out of 28 simulations, handling complex scenarios like global mean interference and time lags. While other baselines excel sporadically, DiffuGC consistently provides robust, high-quality results, demonstrating its superior generalization and resilience. This confirms DiffuGC's practical utility in brain connectivity inference.

CausalTime For the new benchmark datasets, we followed the experimental settings described by [28]. Table IV details the performance of various methods across three quasi-realistic domains from the CausalTime benchmark. DiffuGC emerges as the top performer across all domains and evaluation criteria, consistently surpassing the baselines. Notably, its superiority is especially pronounced in complex domains characterized by distribution shifts and diverse interventions. These findings

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TABLE I: Comparisons of AUROC and AUPRC for Granger causality among different approaches on VAR dataset.

VAR	Metrics	GC	PCMCI	NGC	DyNoTears	CR-VAE	CUTS	IGC	**DiffuGC**
D=10, T=1000, Lag=5	AUROC (↑)	0.687(±0.071)	0.888(±0.035)	0.925(±0.011)	0.806(±0.020)	0.955(±0.001)	0.925(±0.007)	**1.000(±0.000)**	**1.000(±0.000)**
	AUPRC(↑)	0.676(±0.015)	0.863(±0.015)	0.921(±0.010)	0.788(±0.006)	0.989(±0.015)	0.927(±0.010)	**1.000(±0.000)**	**1.000(±0.000)**
	F1(↑)	0.650(±0.010)	0.789(±0.025)	0.917(±0.018)	0.750(±0.015)	0.930(±0.009)	0.920(±0.005)	**1.000(±0.000)**	**1.000(±0.000)**
	SHD(↓)	8(±2)	4(±2)	2(±1)	1(±0)	1(±0)	2(±0)	**0(±0)**	**0(±0)**
D=20, T=1000, Lag=5	AUROC (↑)	0.598(±0.020)	0.666(±0.020)	0.833(±0.025)	0.759(±0.020)	0.925(±0.015)	0.947(±0.010)	0.970(±0.019)	**0.982±0.033**
	AUPRC (↑)	0.605(±0.014)	0.743(±0.017)	0.840(±0.015)	0.745(±0.015)	0.965(±0.018)	0.980(±0.032)	0.970(±0.020)	**0.989(±0.018)**
	F1(↑)	0.599(±0.031)	0.701(±0.028)	0.852(±0.017)	0.733(±0.015)	0.930(±0.024)	0.963(±0.015)	0.969(±0.017)	**0.980(±0.017)**
	SHD(↓)	38(±4)	33(±2)	15(±3)	14(±3)	8(±2)	6(±2)	11(±1)	**5(±1)**
D=20, T=500, Lag=20	AUROC (↑)	0.569(±0.020)	0.598(±0.030)	0.820(±0.020)	0.698(±0.010)	0.830(±0.013)	0.842(±0.025)	0.935(±0.015)	**0.971(±0.012)**
	AUPRC (↑)	0.588(±0.025)	0.587(±0.030)	0.810(±0.021)	0.675(±0.025)	0.835(±0.015)	0.837(±0.027)	0.925(±0.016)	**0.973(±0.006)**
	F1(↑)	0.708(±0.018)	0.609(±0.045)	0.823(±0.005)	0.678(±0.002)	0.810(±0.025)	0.823(±0.015)	0.946(±0.012)	**0.955(±0.002)**
	SHD(↓)	179(±4)	165(±12)	65(±15)	95(±6)	99(±5)	22(±3)	59(±7)	**18(±2)**
D=30, T=1000, Lag=20	AUROC (↑)	0.599(±0.029)	0.566(±0.025)	0.789(±0.015)	0.649(±0.015)	0.785(±0.026)	0.838(±0.020)	0.919(±0.003)	**0.945(±0.005)**
	AUPRC (↑)	0.578(±0.020)	0.589(±0.024)	0.813(±0.021)	0.643(±0.017)	0.845(±0.015)	0.837(±0.017)	0.902(±0.018)	**0.956(±0.010)**
	F1(↑)	0.710(±0.010)	0.578(±0.032)	0.790(±0.010)	0.638(±0.022)	0.805(±0.015)	0.810(±0.029)	0.938(±0.009)	**0.950(±0.002)**
	SHD(↓)	164(±3)	158(±10)	58(±9)	85(±2)	83(±10)	79(±6)	33(±6)	**10(±2)**

TABLE II: Comparisons of AUROC and AUPRC for Granger causality among different approaches on Lorenz-96 dataset.

Lorenz-96	Metrics	GC	PCMCI	NGC	DyNoTears	CR-VAE	CUTS	IGC	**DiffuGC**
D=10, T=1000, F=10	AUROC (↑)	0.658(±0.040)	0.598(±0.045)	0.938(±0.001)	0.765(±0.020)	0.928(±0.040)	0.928(±0.030)	**1.000(±0.000)**	**1.000(±0.000)**
	AUPRC(↑)	0.588(±0.033)	0.601(±0.005)	0.945(±0.008)	0.785(±0.015)	0.922(±0.005)	0.916(±0.017)	**1.000(±0.000)**	**1.000(±0.000)**
	F1(↑)	0.601(±0.019)	0.565(±0.018)	0.930(±0.001)	0.734(±0.016)	0.922(±0.027)	0.905(±0.006)	**1.000(±0.000)**	**1.000(±0.000)**
	SHD(↓)	17(±2)	15(±1)	6(±0)	10(±1)	7(±1)	2(±0)	7(±0)	**3(±0)**
D=20, T=1000, F=10	AUROC (↑)	0.573(±0.026)	0.608(±0.022)	0.862(±0.018)	0.713(±0.020)	0.923(±0.013)	0.850(±0.020)	0.934(±0.018)	**0.979±0.033**
	AUPRC (↑)	0.560(±0.014)	0.634(±0.015)	0.875(±0.009)	0.715(±0.028)	0.893(±0.020)	0.867(±0.021)	0.946(±0.015)	**0.984(±0.018)**
	F1(↑)	0.686(±0.024)	0.635(±0.010)	0.873(±0.021)	0.728(±0.022)	0.903(±0.006)	0.822(±0.019)	0.931(±0.011)	**0.980(±0.017)**
	SHD(↓)	48(±2)	42(±2)	12(±2)	29(±3)	9(±1)	14(±3)	10(±2)	**8(±1)**
D=20, T=500, F=20	AUROC (↑)	0.540(±0.018)	0.575(±0.015)	0.775(±0.016)	0.656(±0.023)	0.853(±0.020)	0.813(±0.038)	0.903(±0.020)	**0.943(±0.008)**
	AUPRC (↑)	0.568(±0.010)	0.586(±0.010)	0.780(±0.014)	0.665(±0.012)	0.867(±0.018)	0.862(±0.017)	0.925(±0.022))	**0.950(±0.015)**
	F1(↑)	0.690(±0.018)	0.571(±0.012)	0.770(±0.010)	0.725(±0.013)	0.565(±0.017)	0.810(±0.026)	0.915(±0.004)	**0.944(±0.006)**
	SHD(↓)	197(±3)	182(±10)	82(±7)	141(±5)	103(±8)	70(±19)	78(±8)	**23(±3)**
D=40, T=1000, F=20	AUROC (↑)	0.560(±0.019)	0.557(±0.013)	0.719(±0.015)	0.716(±0.018)	0.743(±0.021)	0.825(±0.006)	0.907(±0.008)	**0.932(±0.005)**
	AUPRC (↑)	0.556(±0.010)	0.543(±0.029)	0.766(±0.011)	0.687(±0.017)	0.809(±0.017)	0.829(±0.005)	0.909(±0.017)	**0.940(±0.016)**
	F1(↑)	0.571(±0.015)	0.568(±0.042)	0.767(±0.029)	0.755(±0.010)	0.768(±0.024)	0.774(±0.017)	0.913(±0.002)	**0.938(±0.009)**
	SHD(↓)	169(±8)	170(±12)	152(±10)	90(±8)	91(±9)	77(±10)	32(±10)	**15(±5)**

demonstrate that DiffuGC effectively addresses real-world time series challenges, establishing it as a highly robust tool for interventional Granger causal discovery.

E. Results on Intervened Synthethic Time Series

We perform controlled interventions on synthetic VAR and Lorenz-96 datasets to study their impact on causal discovery from interventional time series. We focus on the duration of interventions and the number of variables affected. To highlight intervention effects, we replace selected variables with Gaussian noise, which isolates and amplifies their impact on causal structure accuracy.

Percentage of intervention time. We intervene on 5%, 10%, 20%, and 50% of the time series duration to evaluate how intervention length affects Granger causal discovery. Figure 5 shows that as intervention increases from 5% to 50%, most baseline methods suffer significant performance decline in AUROC and AUPRC due to noise sensitivity. In contrast, DiffuGC consistently achieves high performance and robustly handles various intervention durations in both linear and nonlinear settings, demonstrating its resilience to prolonged interventions and capability in heterogeneous TS environments.

Percentage of intervention variables. We randomly intervene on 10%, 20%, 50%, 100% of the time series to analyze how the proportion of intervention variables affects Granger causal discovery under interventional settings. Figure 6 shows the performance of various causal discovery methods under increasing percentages of intervention variables. As the proportion of intervened variables grows from 10% to 100%, most baseline methods suffer noticeable declines in F1 Score and increases in SHD, indicating a reduction in precision and structural accuracy. In contrast, DiffuGC maintains both high F1 and consistently low SHD across all settings, demonstrating its robustness in the presence of large-scale interventions. This resilience highlights DiffuGC's ability to integrate heterogeneous interventional signals without being misled by extensive structural perturbations, which often degrade the performance of other approaches.

F. Results on Real-World fMRI Dataset

We validate our model by comparing it to alternative methods using a real resting-state fMRI [1] dataset from the enhanced NKI Rockland sample [37] (Fig. 7). The dataset offers voxel-level BOLD signals, grouped into regions of interest (ROIs) via anatomical parcellation. We examine seven ROIs: posterior and anterior cingulate cortexes, middle temporal and angular gyri in both hemispheres (PCC, LACC, LMTG, LAG, RACC,

[1]https://nilearn.github.io/stable/index.html

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TABLE III: AUROC of the fMRI bold signals dataset

Subject	AUROC							
	GC	NGC	PCMCI	DyNoTears	CR-VAE	CUTS	IGC	**DiffuGC**
Sim1	0.579±0.045	0.715±0.032	0.749±0.053	0.18±0.046	0.801±0.048	0.813±0.046	**0.818**±0.032	0.803±0.064
Sim2	0.630±0.038	0.739±0.042	0.709±0.032	0.681±0.028	0.831±0.026	0.848±0.034	0.826±0.038	**0.846**±0.036
Sim3	0.516±0.027	0.724±0.056	0.728±0.048	0.698±0.034	0.848±0.021	0.850±0.026	0.825±0.037	**0.868**±0.027
Sim4	0.602±0.047	0.708±0.025	0.727±0.047	0.676±0.042	0.849±0.015	0.864±0.028	0.868±0.020	**0.901**±0.016
Sim5	0.585±0.035	0.747±0.035	0.729±0.032	0.785±0.026	0.828±0.030	0.845±0.042	0.839±0.045	**0.855**±0.037
Sim6	0.611±0.092	0.766±0.027	0.809±0.033	0.829±0.037	0.877±0.036	0.898±0.034	0.888±0.028	**0.907**±0.025
Sim7	0.617±0.043	0.775±0.045	0.740±0.028	0.795±0.038	0.837±0.036	0.848±0.047	0.831±0.037	**0.895**±0.038
Sim8	0.518±0.090	0.602±0.087	0.628±0.066	0.597±0.113	0.613±0.083	0.677±0.065	0.702±0.055	**0.753**±0.067
Sim9	0.601±0.085	0.701±0.043	0.706±0.066	0.708±0.080	0.732±0.077	0.805±0.062	0.802±0.068	**0.811**±0.077
Sim10	0.640±0.080	0.665±0.109	0.680±0.072	0.723±0.120	0.672±0.070	**0.781**±0.065	0.770±0.085	0.772±0.070
Sim11	0.608±0.037	0.725±0.038	0.738±0.027	0.764±0.037	0.802±0.028	0.808±0.023	0.815±0.026	**0.821**±0.024
Sim12	0.545±0.033	0.737±0.037	0.726±0.040	0.777±0.031	0.788±0.046	0.805±0.040	0.822±0.042	**0.850**±0.030
Sim13	0.580±0.042	0.648±0.081	0.670±0.087	0.676±0.081	0.680±0.090	0.704±0.066	0.728±0.070	**0.751**±0.074
Sim14	0.539±0.067	0.681±0.070	0.664±0.083	0.708±0.070	0.735±0.062	0.750±0.071	0.755±0.060	**0.792**±0.072
Sim15	0.633±0.099	0.600±0.088	0.599±0.075	0.655±0.065	0.688±0.085	0.726±0.080	**0.764**±0.091	0.740±0.075
Sim16	0.603±0.122	0.644±0.086	0.625±0.080	0.633±0.086	0.622±0.118	0.711±0.080	0.700±0.118	**0.744**±0.088
Sim17	0.578±0.053	0.703±0.042	0.687±0.043	0.768±0.038	0.781±0.040	0.839±0.030	0.855±0.044	**0.878**±0.032
Sim18	0.555±0.076	0.674±0.052	0.680±0.062	0.736±0.050	0.743±0.057	0.818±0.051	**0.821**±0.052	0.807±0.062
Sim19	0.566±0.041	0.788±0.050	0.722±0.061	0.819±0.042	0.833±0.040	0.867±0.031	0.860±0.032	**0.892**±0.032
Sim20	0.583±0.091	0.814±0.032	0.750±0.057	0.840±0.043	0.865±0.022	0.902±0.032	0.880±0.015	**0.909**±0.037
Sim21	0.563±0.082	0.689±0.057	0.700±0.043	0.688±0.049	0.632±0.067	0.770±0.060	0.758±0.055	**0.801**±0.060
Sim22	0.578±0.050	0.703±0.057	0.712±0.050	0.658±0.063	0.654±0.058	0.788±0.050	0.789±0.050	**0.818**±0.060
Sim23	0.588±0.092	0.609±0.081	0.614±0.082	0.572±0.083	0.613±0.094	0.635±0.086	**0.696**±0.076	0.657±0.077
Sim24	0.530±0.120	0.545±0.061	0.551±0.081	0.521±0.078	0.585±0.12	**0.605**±0.072	0.565±0.073	0.580±0.092
Sim25	0.537±0.050	0.624±0.066	0.638±0.058	0.600±0.060	0.637±0.040	0.700±0.062	0.713±0.053	**0.757**±0.058
Sim26	0.579±0.089	0.601±0.055	0.620±0.077	0.585±0.053	0.619±0.049	0.677±0.089	0.695±0.072	**0.708**±0.093
Sim27	0.626±0.062	0.638±0.090	0.682±0.062	0.607±0.071	0.639±0.080	0.699±0.069	0.713±0.061	**0.745**±0.082
Sim28	0.644±0.052	0.700±0.050	0.728±0.045	0.631±0.040	0.732±0.072	0.755±0.083	0.760±0.060	**0.810**±0.072

TABLE IV: Comparative performance on CausalTime benchmark datasets.

Methods	Metrics	AQI	Traffic	Medical
GC	AUROC	0.447±0.014	0.451±0.026	0.531±0.041
	AUPRC	0.645±0.027	0.278±0.022	0.421±0.028
CUTS	AUROC	0.624±0.0434	0.618±0.025	0.368±0.021
	AUPRC	0.553±0.036	0.352±0.022	0.353±0.014
NGC	AUROC	0.717±0.019	0.603±0.025	0.574±0.014
	AUPRC	0.717±0.020	0.358±0.049	0.463±0.012
CR-VAE	AUROC	0.504±0.025	0.524±0.041	0.524±0.013
	AUPRC	0.338±0.030	0.421±0.025	0.490±0.021
PCMCI	AUROC	0.426±0.044	0.536±0.016	0.665±0.021
	AUPRC	0.669±0.015	0.392±0.022	0.548±0.024
IGC	AUROC	0.784±0.021	0.619±0.045	0.739±0.012
	AUPRC	0.754±0.012	0.564±0.017	0.542±0.050
DiffuGC	AUROC	**0.805±0.014**	**0.708±0.010**	**0.772±0.011**
	AUPRC	**0.786±0.024**	**0.607±0.023**	**0.656±0.016**

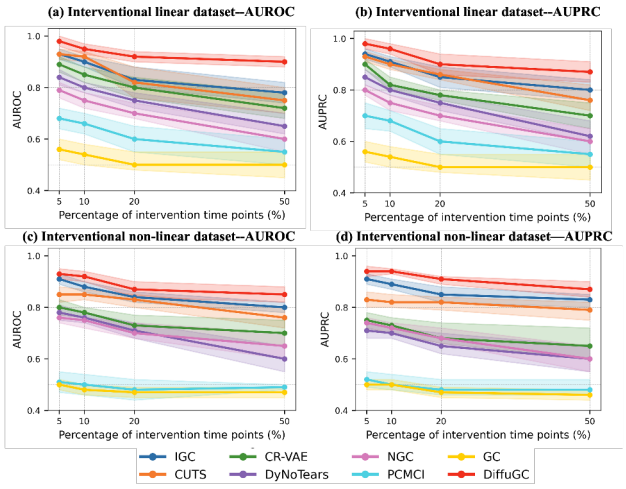


Fig. 5: Comparing causal discovery methods with varied intervention points (5%, 10%, 20%, 50%) on synthetic linear (a,b) and nonlinear (c,d) datasets.

RMTG, RAG). Some regions correlate during rest. For each ROI, voxel signals are averaged to form a representative time series for causal discovery.

According to Fig. 7, DiffuGC identifies the PCC as the main cortical hub, causally influencing the AG, MTG, and LACC, which is consistent with existing neuroscientific studies in [35]. Moreover, the detected causal path MTG→AG aligns with previously reported interactions in [36]. Compared to the state-of-the-art baseline IGC for interventional Granger causal discovery, it fails to uncover the core causal role of PCC.

G. Ablation study

We compare two intrinsic noise schedules for diffusive interventions: linear and cosine β_n. As shown in Fig. 8(a), the linear schedule injects stronger noise early, while the

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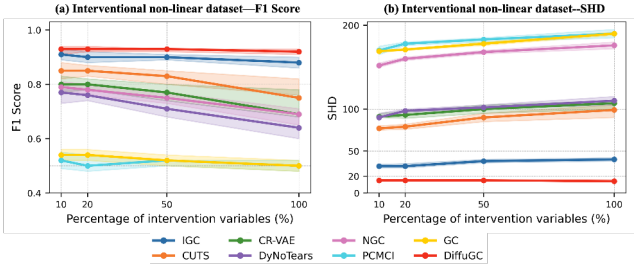


Fig. 6: Performance of causal discovery methods with varying percentages of intervention variables on synthetic datasets using (a) F1 Score and (b) SHD.

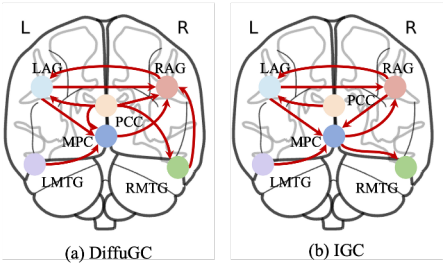


Fig. 7: Causal graphs of 6 selected ROIs estimated by our method (a) and baseline IGC (b) from real-world fMRI dataset.

cosine schedule introduces weaker initial perturbations that increase gradually. This difference shapes the intervention trajectories: Figs. 8(b)–(c) show that linear scheduling perturbs early dynamics more aggressively, whereas cosine scheduling preserves short-term patterns and applies smoother transitions across diffusion steps. Performance comparisons on three datasets are reported in Figs. 8(d)–(e). On synthetic datasets (VAR and Lorenz-96), both schedules perform comparably, confirming that DiffuGC remains robust regardless of the chosen strategy. On the more complex fMRI dataset, however, the cosine schedule achieves slightly better results, suggesting that its gradual noise injection better preserves temporal dependencies under high noise.

H. Case Study–Causality Acceleration

Our experiments reveal a fascinating phenomenon. Figure 9 shows how causal strength changes through diffusion steps for both truly causal and non-causal variable pairs. Significantly, we find that the distinction between causal and non-causal relationships appears early in the diffusion process—true causal edges maintain high strength initially and diminish slowly, while non-causal connections stay weak throughout. This observation leads us to introduce a novel concept: Causality Acceleration, denoting the speed at which causal signals become apparent through successive noisy interventions. Essentially, insightful causal patterns aren't spread uniformly across the entire diffusion path; instead, they become evident more quickly during the initial to middle diffusion stages. This realization suggests new optimization opportunities: by

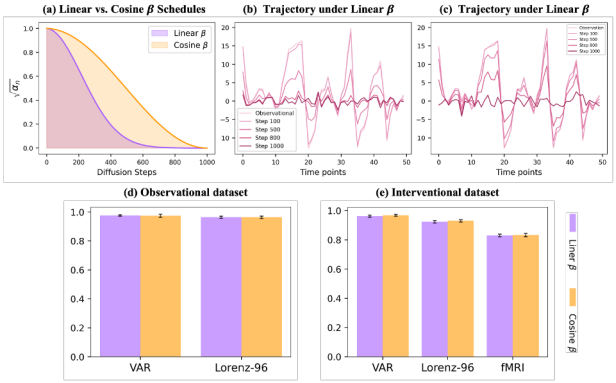


Fig. 8: Comparison of linear and cosine noise schedules in diffusive interventions.

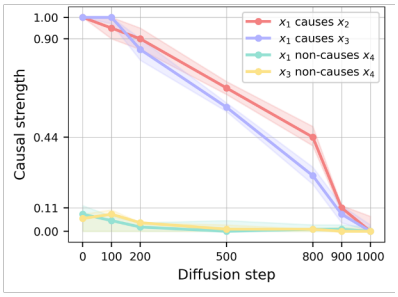


Fig. 9: Causality Acceleration—evolution of causal strength across diffusion steps.

directing computational effort or implementing early stopping at the most significant steps, we may reduce training costs while maintaining precision.

VI. Conclusion

In this work, we introduced DiffuGC, a diffusion-based framework for Granger causal discovery that integrates observational and interventional time series. By coupling progressive diffusive interventions with the Transformer-based denoiser NoiFormer, DiffuGC reliably reconstructs causal structures under noise, heterogeneity, and distribution shifts. Extensive experiments on synthetic, quasi-real, and real-world datasets demonstrate consistent improvements over state-of-the-art baselines in accuracy, robustness, and generalization. Moreover, we identified a notable phenomenon, termed Causality Acceleration, where critical causal signals emerge early in the diffusion process, providing new insights into the dynamics of diffusion-based inference. These findings highlight both the empirical effectiveness and the theoretical potential of DiffuGC. Overall, our study establishes DiffuGC as a principled and adaptable framework for reliable causal discovery in complex time series, and points to promising future directions in efficient, data-driven causal inference.

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VII. Acknowledgements

This work is supported by the National Natural Science Foundation of China (No.62172018, No.62102008) and CCF-Zhipu Large Model Innovation Fund (CCF-Zhipu202414).

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